PVP 14

Code: CE1T1, ME1T1, CS1T1, IT1T1, EE1T1, EC1T1, AE1T1

I B.Tech - I Semester – Regular Examinations - November 2015

ENGINEERING MATHEMATICS - I
(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1. a) Show that the equation $x\,dx + y\,dy = \dfrac{a^2(xdy - ydx)}{x^2 + y^2}$ is exact.

b) Solve the differential equation

$(D^4 - D^3 - 9D^2 - 11D - 4)y = 0$.

c) Find the orthogonal trajectories of family of parabolas

$y = ax^2$.

d) State Rolle's theorem.

e) Write Taylor's series of $\cos x$ in powers of $\left(x - \dfrac{\pi}{2}\right)$

f) Change the order of integration of the double integration

$$\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dy dx$$

g) Evaluate $\displaystyle\int_0^{\pi} \int_0^{x} \sin y \, dy dx$

h) If $f(x,y,z)=3x^2y-y^3z^2$ then find grad f at the point (1, -2, -1).

i) State Gauss divergence theorem.

j) Write the normal equations to fit the parabola $y=a+bx+cx^2$

k) Show that $\beta(m,n)=\beta(m+1,n)+\beta(m,n+1)$.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 16 x 3 = 48 M

2. a) If the air is maintained at 30^0C and the temperature of the body cools from 80^0C to 60^0C in 12 minutes, find the temperature of the body after 24 minutes from the original. 8 M

b) Solve 8 M

$$(D^2-2D+1)y=xe^x\sin x$$

3. a) Using Lagrange's Mean value theorem, show that 8 M

$$x>\log(1+x)>\frac{x}{1+x} \text{ if } f(x)=\log(1+x), \forall x>0.$$

b) Find the dimensions of the rectangular box, open at the top, of maximum capacity whose surface area is 432 sq.cm. 8 M

4. a) Evaluate the integral $\int_0^a \int_y^a \frac{x^2}{\sqrt{x^2+y^2}}\,dxdy$

using the change of order of integration. 8 M

b) Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dxdydz}{\sqrt{1-x^2-y^2-z^2}}$

by changing to spherical co-ordinates. 8 M

5. a) Find the directional derivative of the function

$f = x^2 - y^2 + 2z^2$ at the point P(1,2,3) in the direction of the line PQ where Q=(5,0,4). 8 M

b) Evaluate $\int_S \overline{F}.\overline{N}\,dS$ where $\overline{F} = z\,\overline{i} + x\,\overline{j} - 3y^2 z\,\overline{k}$ and S is the surface $x^2 + y^2 = 16$ included in the first octant between z=0 and z=5. 8 M

6. a) Fit a least square geometric curve $y = ax^b$ to the following data. 8 M

x:	1	2	3	4	5
y:	0.5	2	4.5	8	12.5

b) Derive the relation between Beta and Gamma function.

8 M

PVP 14

Code: CE1T1, ME1T1, CS1T1, IT1T1, EE1T1, EC1T1, AE1T1

I B. Tech - I Semester – Regular / Supplementary Examinations
December 2016

ENGINEERING MATHEMATICS - I
(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1.

a) Write any two Integrating Factors and when they are to be used to convert a non exact differential equation to exact differential equation?

b) Obtain the 1st order differential equation that is applicable to Newton's law of cooling.

c) Find the Particular Integral of $\left(D^3+1\right)y = Cosx$.

d) Write the Meclaurin's series for the function f(x) in powers of (x-2) upto 3rd degree.

e) What is the nature of Stationary points and Saddle points of a function?

f) What is the area of the circle r = 2 in the form of a double integral in Cartesian coordinates?

g) Evaluate $\iint_R xy\,dx\,dy$ where R is bounded by $x = 0$, $y = 0$, $x = a$, $y = a$.

h) What are Solenoidal and Irrotational vectors?

i) Find the unit normal vector to the surface $x^2y = 3z$ at (1,3,1).

j) Obtain the value of $\Gamma(25/2)$.

k) What is the significance of least squares method of fitting a curve?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) Find the orthogonal trajectories of confocal conics,

$$\frac{x^2}{a^2} + \frac{y^2}{a^2+\lambda} = 1, \quad \text{where } \lambda \text{ is the parameter.}$$ 6 M

b) Solve the differential equation $\frac{d^2y}{dx^2} + a^2 y = \operatorname{Sec} ax$. 10 M

3. a) Find the minimum value of $x^2 + y^2 + z^2$ given that $ax + by + cz = p$. 8 M

b) Show that by Taylor's theorem, with lagranges form of reminder for any x,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\ldots + (-1)^{n-1}\frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n}}{(2n)!}\sin\theta x, \quad 0 < \theta < 1$$

8 M

4. a) Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dx\, dy\, dz$. 8 M

b) By changing to polar coordinates, evaluate $\iint \frac{x^2 y^2}{x^2+y^2} dx\, dy$ over the annular region between the circles $x^2+y^2=a^2, x^2+y^2=b^2, b>a$. 8 M

5. a) Find the constants a, b, c so that the vector

$\bar{A} = (x+2y+az)\bar{i} + (bx-2y-z)\bar{j} + (4x+cy+2z)\bar{k}$

is irrotational. Also find ϕ such that $\bar{A} = \nabla \phi$. 8 M

b) State Stoke's theorem and hence or otherwise evaluate $\int [(x+y)dx + (2x-z)dy + (y-z)dz]$ over the closed boundary of the triangle with vertices (0,0,0), (1,0,0) and (1,1,0). 8 M

6. a) Determine the constants 'a' and 'b' by the least squares method such that $y = ae^{bx}$ fits the following data. 9 M

x	1	1.2	1.4	1.6
y	40.17	73.196	133.372	243.02

b) Show that $\Gamma(1/2) = \sqrt{\pi}$ using β and Γ integrals. 7 M

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I B. Tech - I Semester – Regular / Supplementary Examinations
November 2017

ENGINEERING MATHEMATICS - I

(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.

a) Solve $(x^2 - ay)dx = (ax - y^2)\,dy$.

b) State Newton's law of cooling.

c) Express e^x as powers of x.

d) State Rolle's theorem.

e) Find the particular integral of $(D^2 - 1)y = e^{2x}$.

f) Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2)\,dy\,dx$.

g) Show that $div\,\vec{r} = 3$

h) Show that $curl(grad\,\varphi) = \bar{0}$.

i) Using Green's theorem, find the area of the ellipse $x = a\cos\theta,\ y = b\sin\theta,\ 0 \le \theta \le 2\pi$.

j) Write the normal equations of a straight line $y = ax + b$.

k) Calculate $\Gamma\left(-\frac{3}{2}\right)$.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2.a) Show that the family of con-focal conics $\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}=1$ is selforthogonal. Where λ is parameter. 8 M

b) Solve: $(D^2-2D+1)y = xe^x \sin x$. 8 M

3.a) Using Lagrange's mean value theorem, prove that

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}.$$

Hence deduce that $\frac{\pi}{4}+\frac{3}{25} < \tan^{-1}\frac{4}{3} < \frac{\pi}{4}+\frac{1}{6}$.

Given that 0<a<b<1. 8 M

b) A rectangular box open at the top is to have volume 32 cubic feet. Find the dimensions of the box requiring least material for its construction. 8 M

4.a) Calculate $\iint r^3 dr d\theta$ over the area included between the circles $r = 2\sin\theta$ and $r = 4\sin\theta$. 8 M

b) Change the order of integration in $I = \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$ and hence evaluate. 8 M

5. a) Find the directional derivative of $\varphi = x^2 y + 4xz^2$ at the point $(1, -2, 1)$ in the direction of the vector $2i - j - 2k$. 8 M

b) Use divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n}\, ds$ where $\vec{F} = 4xi - 2y^2 j + z^2 k$ and **'S'** is the surface of the cylinder $x^2 + y^2 = 4, Z = 0 \; and \; Z = 3$ 8 M

6. a) By the method of least squares find the equation of the straight line that best fits the following data. 8 M

x	1	2	3	4	5
Y	14	27	40	55	68

b) Define Beta and Gamma functions, then show that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. 8 M

PVP 14

Code: CE1T1, ME1T1, CS1T1, IT1T1, EE1T1, EC1T1, AE1T1

I B. Tech - I Semester – Regular / Supplementary Examinations
November 2018

ENGINEERING MATHEMATICS - I

(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.

a) If $\frac{dx}{dt} + \frac{x}{t \log t} = \frac{1}{t}$, then find the integrating factor.

b) Find the particular integral of $(D^2+1)y = \sin x$

c) State Newton's law of cooling.

d) State Rolle's theorem.

e) Expand e^x in powers of x second degree term.

f) Evaluate $\int_{\theta=0}^{\frac{\pi}{2}} \int_{r=0}^{\infty} e^{-r^2} r dr d\theta$

g) Evaluate $\int_0^1 \int_y^1 e^{-x} \; dx \, dy$

h) State Stoke's theorem.

i) Find $div\, \bar{F} \; if \; \bar{F} = xyz\boldsymbol{i} + 3x^2y\boldsymbol{j} + (xz^2 - y^2z)\boldsymbol{k}$ at point (2,-1,1).

j) Show that $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} sin^{2m-1} x \, cos^{2n-1} x \, dx$

k) Write normal equations to fit a quadratic curve $y=ax^2+bx+c$

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) Solve $(D^2+1)y = \sec x$ by method of parameters. 8 M

b) Show that the family of confocal and coaxial parabolas $y^2 = 4a(x+a)$ where a is an arbitrary constant are self orthogonal. 8 M

3. a) In the plane triangle ABC, find the maximum value of $\cos A \cos B \cos C$ 8 M

b) Verify Rolle's Theorem for the function $f(x) = \frac{x^2-4x}{x+2}$ in (0,4) 8 M

4. a) Evaluate $\iint (x+y)^2\, dxdy$ over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 8 M

b) Evaluate the double integral $\int_0^a \int_{\sqrt{ax}}^a \frac{y^2}{\sqrt{y^4-a^2x^2}}\, dxdy$ by changing the order of integration. 8 M

5. a) Find the directional derivative of the function $f(x,y,z) = x^2+y^2+2z^2$ at the point $P(-1,2,3)$ in the direction of $4i-2j+k$. 8 M

b) Using Green's theorem, evaluate $\int_C (3x^2 - 8y^2)\, dx + (4y - 6xy) dy$ where C is a boundary of the region bounded by $y = \sqrt{x}\ and\ y = x^2$. 8 M

6. a) Define Beta function and gamma function and find the relation between them. 8 M

b) Fit a second degree polynomial for the data $(1,2), (2,5), (4,9), (6,10)$. 8 M